## Title of the Practical

**Transfer Learning and Fine-Tuning using a Pretrained Convolutional Neural Network (CNN)**

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## Aim / Objective

- To implement **Transfer Learning** using a state-of-the-art pretrained CNN architecture (**ResNet50** or **VGG16**).
- To adapt and fine-tune the pretrained model on a small, custom binary dataset (**Cats vs. Dogs**).
- To experiment with **layer freezing** and evaluate its subsequent impact on model accuracy and training efficiency.

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## Theory & Concept Background

Training a deep CNN from scratch requires thousands of labeled images and massive computational power. **Transfer Learning** bypasses this constraint by taking a model already trained on a large dataset (like **ImageNet**) and reusing its learned features for a new, related task.

A standard CNN operates in two main segments:

1. **Convolutional Base (Feature Extractor):** The early layers extract generic features like edges and textures, while deeper layers extract high-level abstract shapes. These weights are **frozen** so their pre-learned knowledge is preserved.
2. **Classifier Head:** The final dense (fully connected) layers map features to specific class targets. This part is removed and replaced with a new custom classifier matching the custom dataset's labels (e.g., 2 outputs for Cats vs. Dogs).

## Hardware & Software Requirements

- **Operating System:** Windows / Linux / macOS
- **Development Environment:** Google Colab / Jupyter Notebook
- **Libraries & Frameworks:** Python, TensorFlow/Keras or PyTorch, NumPy, Matplotlib

## Algorithm: Transfer Learning with a Pretrained CNN

- **Input:** A custom dataset of labeled images (e.g., Cats and Dogs), split into training and validation sets.
- **Output:** A trained and optimized deep learning model capable of binary image classification.

**Phase 1: Data Preparation & Initialization**

1. **Load and Split Dataset:** Acquire the custom dataset and split it into discrete Training and Validation subsets.
2. **Data Preprocessing:**

- Resize all raw input images to match the strict architectural input specifications of the chosen base network (e.g., **224 × 224 × 3** dimensions).
- Normalize image pixel values from the standard \([0, 255]\) range down to an optimized \([0, 1]\) array.

3. **Data Augmentation:** Apply random spatial transformations (such as rotations, zooms, and horizontal flips) exclusively to the training subset to structurally prevent overfitting.

**Phase 2: Network Construction**

4. **Load Pretrained Architecture:** Instantiate the baseline network architecture (**ResNet50** or **VGG16**) embedded with weights previously optimized on the **ImageNet** dataset.
5. **Sever Top Classification Layers:** Explicitly remove the default, final fully connected classification layer (include_top=False) from the imported network architecture.
6. **Append Custom Classifier Head:** Stack a new, structurally custom classification sequence onto the open output of the pretrained base:
   - Add a **Global Average Pooling** layer to flatten the extracted feature maps.
   - Add a **Dense (Fully Connected)** layer with a ReLU activation function and an integrated Dropout layer to enforce regularization.
   - Add a final **Dense Output** layer with a Sigmoid activation function configured for binary targets.

**Phase 3: Experiment A – Feature Extraction (Frozen Base)**

7. **Freeze the Feature Extractor:** Set the training parameters of the entire pretrained base network to non-trainable (trainable = False).
8. **Compile & Train Model:**
   - Compile the network using the **Adam optimizer** (with a standard learning rate, e.g., $\eta = 0.001$) and **Binary Cross-Entropy loss**.
   - Execute model training over a fixed number of epochs (N).
9. **Log Output Metrics:** Compute and archive the baseline training accuracy, validation accuracy, and execution runtime per epoch.

**Phase 4: Experiment B – Fine-Tuning (Partial Unfreezing)**

10. **Unfreeze Targeted Deeper Layers:** Transition the pretrained network base to trainable, but explicitly freeze the early structural layers (e.g., layers 1 through 15), leaving only the final convolutional block completely open for parameter updates.
11. **Recompile Model with Micro-Learning Rate:** Recompile the entire network structure using a heavily reduced learning rate (e.g., $\eta = 0.00001$) to prevent damaging the critical structural weights already present in the network.
12. **Retrain Model:** Execute the training loop for an additional sequence of epochs (M) to allow the unfrozen convolutional block to naturally align with the features of the new custom dataset.

**Phase 5: Evaluation**

13. **Comparative Evaluation:** Extract, plot, and directly contrast the loss and accuracy metrics recorded during both Phase 3 and Phase 4 to assess performance gains.

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**conclusion**

This assignment demonstrates the massive utility of **Transfer Learning** when working with limited labeled data. Freezing the base layers acts as an efficient starting point, yielding high computational performance and solid accuracies quickly. Selectively unfreezing deeper layers for fine-tuning refines the model's domain adaptation capabilities, boosting accuracy further at the expense of longer training iterations.